

CURRICULUM SPECIFICATIONS
PHYSICS 2 (SP025)

1.	Name and code of course:	Physics 2 (SP025)
2.	Synopsis:	This course is offered to students to acquire concepts in physics as well as to enhance students' manipulative skills through practical sessions.
3.	Name(s) of academic staff:	Aifaa binti Awang Kechik
4.	Semester offered:	Semester II
5.	Credit value:	5
6.	Prerequisite/ co-requisite (if any):	None
7.	Course Learning Outcomes (CLO): At the end of the course, students should be able to: 1. Explain basic concepts of electricity, magnetism, optics and modern physics. (C2, PLO 1, MQF LOC i) 2. Solve problems of electricity, magnetism, optics and modern physics. (C4, PLO 2, MQF LOC ii) 3. Apply the appropriate scientific laboratory skills in physics experiments. (P3, PLO 3, MQF LOC iii a) 4. Interpret and use familiar and uncomplicated numerical and graphical data to solve problems in physics. (C4, PLO 7, MQF LOC iii e) <i>Note: The MQF 2.0 learning outcomes clusters (LOC) are:</i> i. Knowledge and understanding ii. Cognitive skills iii. Functional work skills with focus on: a. Practical skills b. Interpersonal skills c. Communication skills d. Digital skills e. Numeracy skills f. Leadership, autonomy and responsibility iv. Personal and entrepreneurial skills v. Ethics and professionalism	

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8.

Mapping of Programme Learning Outcomes (PLO) to Programme Educational Objectives (PEO):

PEO PLO	PEO 1	PEO 2	PEO 3	PEO 4	PEO 5
PLO 1	✓				
PLO 2		✓			
PLO 3	✓				
PLO 4			✓		
PLO 5			✓		
PLO 6				✓	
PLO 7		✓			
PLO 8					✓
PLO 9				✓	
PLO 10					✓

Programme Educational Objectives (PEO)

Upon a year of graduation from the programme, graduates are:

PEO 1

Knowledgeable and technically competent in their science disciplines in-line with higher educational institution requirement.

PEO 2

Able to apply information and use data to solve problems in science disciplines.

PEO 3

Able to communicate competently and collaborate effectively in group work to compete in higher education environment.

PEO 4

Able to use basic information technologies and engage in life-long learning to continue the acquisition of new knowledge and skills.

PEO 5

Able to demonstrate leadership skills and practice good values and ethics in managing organisations.

Programme Learning Outcomes (PLO)

At the end of the programme, students should be able to:

PLO 1

Acquire knowledge of science and mathematics as a fundamental of higher level education.

PLO 2

Apply logical, analytical and critical thinking in scientific studies and problem solving.

PLO 3

Demonstrate manipulative skills in laboratory works.

PLO 4

Collaborate in group work with skills required for higher education.

PLO 5

Deliver ideas, information, problems and solution in verbal and written communication.

PLO 6

Use basic digital technology to seek and analyse data for management of information.

PLO 7

Interpret familiar and uncomplicated numerical data to solve problems.

PLO 8

Demonstrate leadership, autonomy and responsibility in managing organization.

PLO 9

Initiate self-improvement through independent learning.

PLO 10

Practice good values, attitude, ethics and accountability in STEM and professionalism.

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9.	Mapping of Course Learning Outcomes (CLO) to Programme Learning Outcomes (PLO), Teaching Methods and Assessment Methods:													
	PLO CLO	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	Teaching Method	Assessment Method	
	CLO 1	✓										Lecture	Summative Assessment Test (UPS)	
	CLO 2		✓									Tutorial	Examination Assignment 1 Lab Report	
	CLO 3			✓								Practical	Practical Test	
	CLO 4							✓				Tutorial	Assignment 2	
	(This description must be read together with Standard 2.1.2, 2.2.1 and 2.2.2 in Area 2)													
10.	Transferable Skills (if applicable):					Critical Thinking and Problem Solving								
11.	Distribution of Student Learning Time (SLT):													
	Course Content Outline						CLO	Learning & Teaching Activities						Total SLT
								Guided Learning / e-Learning (F2F)				Guided Learning (NF2F)	Independent Learning (NF2F)	
								L	T	P	O			
	1.0 Electrostatics 1.1 Coulomb’s Law 1.2 Electric field 1.3 Electric potential 1.4 Charge in a uniform electric field						1, 2	2	6	0	0	0	8	16
	2.0 Capacitors and Dielectrics 2.1 Capacitance and capacitors in series and parallel 2.2 Charging and discharging capacitors 2.3 Capacitors with dielectrics						1, 2, 3	2	6	2	0	0	8	18
	3.0 Electric Current And Direct-Current Circuits 3.1 Electrical current 3.2 Ohm’s Law and resistivity 3.3 Variation of resistance with temperature 3.4 Electromotive force (emf), internal resistance and potential difference 3.5 Resistors in series and parallel 3.6 Kirchhoff’s Rules 3.7 Electrical energy and power 3.8 Potential divider 3.9 Potentiometer						1, 2, 3	2	12	4	0	0	14	32

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4.0	Magnetism	1, 2, 3, 4	2	8	2	0	0	10	22
4.1	Magnetic field								
4.2	Resultant magnetic field produced by current-carrying conductor								
4.3	Force on a moving charged particle in a uniform magnetic field								
4.4	Force on a current carrying conductor in a uniform magnetic field								
4.5	Forces between two parallel current-carrying conductors								
4.6	Application of motion of charged particle								
5.0	Electromagnetic Induction	1, 2	2	6	0	0	0	8	16
5.1	Magnetic flux								
5.2	Induced emf								
5.3	Self-inductance								
5.4	Energy stored in inductor								
5.5	Mutual inductance								
6.0	Alternating Current	1, 2	2	6	0	0	0	8	16
6.1	Alternating current								
6.2	Root mean square (rms)								
6.3	Resistance, reactance and impedance								
6.4	Power and power factor								
7.0	Optics	1, 2, 3	3	16	4	0	0	19	42
7.1	Reflection at a spherical surface								
7.2	Refraction at a spherical surface								
7.3	Thin lenses								
7.4	Huygen's Principle								
7.5	Constructive and destructive interferences								
7.6	Interference of transmitted light through double-slits								
7.7	Interference of reflected light in thin films								
7.8	Diffraction by a single slit								
7.9	Diffraction grating								
8.0	Wave Properties of Particle	1, 2	1	2	0	0	0	3	6
8.1	de Broglie wavelength								
8.2	Electron diffraction								
9.0	Nuclear and Particle Physics	1, 2	2	4	0	0	0	6	12
9.1	Binding energy and mass defect								
9.2	Radioactivity								
9.3	Particle accelerator								
9.4	Fundamental particle								
Total			18	66	12	0	0	84	180

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	Continuous Assessment	Percentage (%)	F2F	NF2F	Total SLT
	1. Assignment (Topic 4 - CTPS)	10	0	2	2
	1. Assignment (Topic 4 - Numeracy)	5	0	1	1
	2. Practical Test	10	1	3	4
	3. Lab Report	15	1	3	4
	Final Assessment	Percentage (%)	F2F	NF2F	Total SLT
	1. Summative Assessment Test	20	1.5	4.5	6
	2. Examination	40	2	6	8
GRAND TOTAL SLT					205
L = Lecture, T = Tutorial, P = Practical, O = Others, F2F = Face-to-Face, NF2F = Non Face-to-Face					
12.	Special requirement or resources:	None			
13.	References:	Cutnell, J. D., Johnson, K. W., Young, D., & Stadler, S. (2019). <i>Introduction to Physics</i> (11th ed.). Hoboken, N.J: John Wiley & Sons. Serway, R. A. & Jewett, J. A. (2019). <i>Physics for Scientists and Engineers</i> (10th ed.). International Student Edition. USA: Brooks/ Cole Cengage Learning. Giancoli, D. C. (2016). <i>Physics – Principles with Application</i> (7th ed.). Prentice Hall.			
14.	Other additional information:	Teaching Learning Outcomes:			

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	Topic 1: Electrostatics 1.1 Coulomb's Law - State Coulomb's Law, $F = \frac{Qq}{4\pi\epsilon_0 r^2} = \frac{kQq}{r^2}$ - Sketch the electric force diagram. - Apply Coulomb's Law for a system of point charges. <i>*simple configuration of charges with a maximum four charges in 2D</i> 1.2 Electric field - Define and use electric field strength, $\vec{E} = \frac{\vec{F}}{q_0}$ - Use $E = \frac{kQ}{r^2}$ for point charge. - Sketch the electric field strength diagram. <i>* simple configuration of charges with a maximum four charges in 2D</i> - Determine the electric field strength E for a system of charges. <i>* simple configuration of charges with a maximum four charges in 2D</i> 1.3 Electric potential - Define electric potential, $V = \frac{W}{q_0}$ - Define and sketch equipotential lines and surfaces of: - an isolated charge; and - a uniform electric field. - Use $V = \frac{kQ}{r}$ for a point charge and a system of charges. <i>* simple configuration of charges with a maximum four charges in 2D</i> - Apply potential difference between two points: $\Delta V = V_{final} - V_{initial}$, $\Delta V = \frac{W}{q_0}$ - Apply the change in potential energy between two points in electric field: $\Delta U = q\Delta V$ - Apply potential energy of a system of point charges: $U = k \left(\frac{q_1 q_2}{r_{12}} + \frac{q_1 q_3}{r_{13}} + \frac{q_2 q_3}{r_{23}} \right)$ up to maximum three charges. 1.4 Charge in a uniform electric field - Analyse the motion of a charge qualitatively and quantitatively in a uniform electric field for each of the following case, - stationary charge; - charge moving perpendicularly to the field; - charge moving parallel to the field; and - charge in dynamic equilibrium. - Use $E = \frac{\Delta V}{d}$ for uniform electric field.
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	Topic 2: Capacitors and Dielectrics 2.1 Capacitance and capacitors in series and parallel - Define and use capacitance, $C = \frac{Q}{V}$ - Determine the effective capacitance of capacitors in series: $\frac{1}{C_{\text{effective}}} = \frac{1}{C_1} + \frac{1}{C_2} + \dots + \frac{1}{C_n}$ and in parallel: $C_{\text{effective}} = C_1 + C_2 + \dots + C_n$ - Apply energy stored in a capacitor, $U = \frac{1}{2}CV^2 = \frac{1}{2}QV = \frac{1}{2}\frac{Q^2}{C}$ 2.2 Charging and discharging of capacitors - State physical meaning of time constant and use $\tau = RC$ - Sketch and explain the characteristics of Q-t and I-t graph for charging and discharging of a capacitor. - Use: $Q = Q_0 e^{\frac{-t}{RC}}$ for discharging; and $Q = Q_0 \left(1 - e^{\frac{-t}{RC}}\right)$ for charging. - Determine the time constant of an RC circuit. (Experiment 1: Capacitor) - Determine the capacitance of a capacitor using an RC circuit. (Experiment 1: Capacitor) 2.3 Capacitors with dielectrics - Define and use dielectric constant, $\epsilon_r = \frac{\epsilon}{\epsilon_0}$ - Describe the effects of dielectric on a parallel plate capacitor. - Apply capacitance of air-filled parallel plate capacitor, $C_0 = \frac{\epsilon_0 A}{d}$ - Determine capacitance with dielectric, $C = \epsilon_r C_0$
	Topic 3: Electric Current and Direct-Current Circuits 3.1 Electrical current - Describe microscopic model of current. <i>*emphasise on the flow of free electrons in a metal. Include concept of drift velocity.</i> - Define electric current, $I = \frac{dQ}{dt}$ - Use electric current, $I = \frac{dQ}{dt}$, $Q = ne$ 3.2 Ohm's Law and resistivity - State and use Ohm's Law. - Define and use resistivity, $\rho = \frac{RA}{l}$ - Sketch V-I graph. (Experiment 2: Ohm's Law) - Verify Ohm's Law. (Experiment 2: Ohm's Law) - Determine effective resistance of resistors in series and parallel by graphing method. (Experiment 2: Ohm's Law)

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3.3 Variation of resistance with temperature a) Explain the effect of temperature on electrical resistance in metals. b) Use $R = R_0[1 + \alpha(T - T_0)]$ <i>* α is at temperature 20°C</i> 3.4 Electromotive force (emf), internal resistance and potential difference a) Define emf, ε and internal resistance, r of a battery. b) State factors that influence the internal resistance. c) Explain the relationship between emf of a battery and potential difference across the battery terminals. d) Use terminal voltage, $V = \varepsilon - Ir$ 3.5 Resistors in series and parallel a) Determine the effective resistance of resistors in series and parallel. 3.6 Kirchhoff's Rules a) State and apply Kirchhoff's Rules. <i>*maximum two closed circuit loops</i> <i>**use calculator to solve the simultaneous equations</i> 3.7 Electrical energy and power a) Use power, $P = IV$, $P = I^2R$ and $P = \frac{V^2}{R}$ (known as power loss). b) Use electrical energy, $W = IVt$ 3.8 Potential divider a) Explain principle of potential divider. b) Use equation of potential divider, $V_1 = \left(\frac{R_1}{R_1 + R_2 + \dots + R_n} \right) V$ <i>*maximum three resistors</i> 3.9 Potentiometer a) Explain principles of potentiometer and its applications. b) Use related equations for potentiometer, $\frac{\varepsilon_1}{\varepsilon_2} = \frac{l_1}{l_2}$ c) Determine internal resistance, r of a dry cell by using potentiometer. (Experiment 3: Potentiometer)	Topic 4: Magnetism 4.1 Magnetic field a) Define magnetic field. b) Identify magnetic field sources. <i>*e.g: Bar magnet & current-carrying conductor (straight wire, circular coil, and solenoid), Earth magnetic field</i> c) Sketch magnetic field lines for: i. bar magnet and current-carrying conductor (straight wire, circular coil, and solenoid); and ii. Earth magnetic field. d) Determine the value of the horizontal component of the earth magnetic field, $\vec{B}_E$. (Experiment 4: Magnetic Field)
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	4.2 Resultant magnetic field produced by current-carrying conductor a) Sketch and determine resultant magnetic field diagram at a point <i>*limited to two current carrying straight wires and 2D</i> b) Determine direction of $\vec{B}$ by using right hand rule. c) Determine the magnitude of magnetic field by using: i. $B = \frac{\mu_0 I}{2\pi r}$ for a long straight wire; ii. $B = \frac{\mu_0 I}{2r}$ at the centre of a circular coil; iii. $B = \mu_0 nI$ at the centre of a solenoid; and iv. $B = \frac{1}{2}\mu_0 nI$ at the end of a solenoid. 4.3 Force on a moving charged particle in a uniform magnetic field a) Explain and use magnetic force, $\vec{F} = q\vec{v} \times \vec{B}$ b) Determine the direction of force. c) Describe circular motion of a charge in a uniform magnetic field. d) Use relationship of magnetic force, $F_B = F_c$ 4.4 Force on a current carrying conductor in a uniform magnetic field a) Explain and use magnetic force, $\vec{F} = I\vec{l} \times \vec{B}$ b) Determine the direction of force 4.5 Forces between two parallel current-carrying conductors a) Explain magnetic force per unit length of two parallel current-carrying conductors. b) Apply magnetic force per unit length, $\frac{F}{l} = \frac{\mu_0 I_1 I_2}{2\pi d}$ 4.6 Application of motion of charged particle a) Explain the motion of a moving charged particle in magnetic field and electric field for v, B and E perpendicular to each other. b) Use velocity, $v = \frac{E}{B}$ in a velocity selector. <i>*e.g: Bainbridge mass spectrometer</i>
	Topic 5: Electromagnetic Induction 5.1 Magnetic flux a) Define and use magnetic flux, $\Phi = \vec{B} \cdot \vec{A} = BA \cos \theta$ b) Use magnetic flux linkage, $\Phi = N\Phi$ 5.2 Induced emf a) Explain induced emf by using Faraday's experiment. b) State and use Faraday's Law, $\varepsilon = -\frac{d\Phi}{dt}$ c) State and use Lenz's Law to determine the direction of induced current. d) Apply induced emf in: i. a straight conductor, $\varepsilon = Blv \sin \theta$ ii. a coil, $\varepsilon = -NA \frac{dB}{dt}$, $\varepsilon = -NA \frac{dA}{dt}$ iii. a rotating coil, $\varepsilon = NAB\omega \sin \omega t$

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	5.3 Self-inductance a) Define self-inductance. b) Apply self-inductance, $L = -\frac{\varepsilon}{\left(\frac{dI}{dt}\right)}$ for coil and solenoid, where: i. $L = \frac{N\Phi}{I}$ ii. $L_{\text{coil}} = \frac{\mu_0 N^2 A}{2r}$ iii. $L_{\text{solenoid}} = \frac{\mu_0 N^2 A}{l}$ 5.4 Energy stored in inductor a) Apply the energy stored in an inductor, $U = \frac{1}{2}LI^2$ 5.5 Mutual inductance a) Define mutual inductance. b) Use mutual inductance, $M = \frac{\mu_0 N_1 N_2 A}{l}$ between two coaxial solenoids.
	Topic 6: Alternating Current 6.1 Alternating current a) Define alternating current (AC). b) Sketch and interpret sinusoidal AC waveform. c) Use sinusoidal voltage and current equations: i. $V = V_o \sin \omega t$ ii. $I = I_o \sin \omega t$ 6.2 Root mean square (rms) a) Define root mean square (rms) current and voltage for AC source. b) Use $I_{\text{rms}} = \frac{I_o}{\sqrt{2}}$, $V_{\text{rms}} = \frac{V_o}{\sqrt{2}}$ 6.3 Resistance, reactance and impedance a) Sketch and use phasor diagram and sinusoidal waveform to show the phase relationship between current and voltage for a single component circuit of: i. resistor, R ii. capacitor, C iii. inductor, L b) Use phasor diagram to analyse voltage, current, and impedance of series circuit of RL, RC and RLC. c) Define and use: i. capacitive reactance, $X_C = \frac{1}{2\pi fC}$ ii. inductive reactance, $X_L = 2\pi fL$ iii. impedance, $Z = \sqrt{R^2 + (X_L - X_C)^2}$ iv. phase angle, $\phi = \tan^{-1} \frac{(X_L - X_C)}{R}$ d) Explain graphically the dependence of R, X_C, X_L and Z on f and relate it to resonance.

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6.4 Power and power factor a) Apply in AC circuit consisting of R, RC, RL and RLC in series: i. average power, $P_{av} = I_{rms} V_{rms} \cos \phi$ <i>*also known as power loss that only occurs in resistor</i> ii. instantaneous power, $P = IV$ iii. power factor, $\cos \phi = \frac{P_r}{P_a} = \frac{P_{av}}{I_{rms} V_{rms}}$	Topic 7: Optics 7.1 Reflection at a spherical surface a) State radius of curvature, $R = 2f$ for spherical mirror. b) Sketch ray diagrams with a minimum of two rays to determine the characteristics of image formed by spherical mirrors. c) Use mirror equation, $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$ for real object only. <i>*sign convention for focal length, f and radius of curvature, R:</i> i. positive f and R for concave mirror; and ii. negative f and R for convex mirror d) Apply magnification, $M = \frac{h_i}{h_o} = -\frac{v}{u}$ 7.2 Refraction at a spherical surface a) Use $\frac{n_1}{u} + \frac{n_2}{v} = \frac{n_2 - n_1}{R}$ for spherical surface. <i>*sign convention for radius of curvature, R:</i> i. positive R for convex surface; and ii. negative R for concave surface 7.3 Thin lenses a) Use thin lens equation, $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$ for real object only. <i>*sign convention for focal length, f:</i> i. positive f for convex lens; and ii. negative f for concave lens b) Determine the focal length of a convex lens. (Experiment 5: Geometrical Optics) c) Use lens maker's equation, $\frac{1}{f} = \left(\frac{n_{\text{material}}}{n_{\text{medium}}} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$ d) Apply magnification, $M = \frac{h_i}{h_o} = -\frac{v}{u}$. e) Use the thin lens formula for a combination of two convex lenses 7.4 Huygens's Principle a) State Huygens's Principle (e.g. spherical and plane wave fronts). b) Sketch and explain the wave front of light after passing through a single slit and obstacle using Huygens's Principle. 7.5 Constructive and destructive interferences a) Define coherence. b) State the conditions for interference of light. c) State the conditions of constructive and destructive interference for in phase and antiphase sources. <i>*emphasise on the path difference and its equivalence to phase difference</i>
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	7.6 Interference of transmitted light through double-slits a) Use: $y_m = \frac{m\lambda D}{d}$ for bright fringes (maxima); and $y_m = \frac{(m+\frac{1}{2})\lambda D}{d}$ for dark fringes (minima), where $m = 0, \pm 1, \pm 2, \pm 3, \dots$ <i>*bright fringes: $m = 0$, central or 0^{th} order max, $m = 1$, first bright or 1^{st} order max</i> <i>*dark fringes: $m = 0$, first dark or 0^{th} order min, $m = 1$, 2^{nd} dark or 1^{st} order min</i> b) Use $\Delta y = \frac{\lambda D}{d}$ and explain the effect of changing any of the variables. <i>*Δy: separation between two consecutive dark or bright fringes</i> 7.7 Interference of reflected light in thin films a) Identify the occurrence of phase change upon reflection. <i>*from lower to higher refractive index, phase change = π rad or path difference = $\frac{1}{2}\lambda$</i> b) Describe with the aid of a diagram the interference of light in thin films at normal incidence. <i>*limited to three media</i> c) Use the following equations for reflected light with no phase difference (non-reflective coating): - Constructive interference: $2nt = m\lambda$ - Destructive interference: $2nt = (m + \frac{1}{2})\lambda$ d) Use the following equations for reflected light of phase difference π rad (reflective coating): - Constructive interference: $2nt = (m + \frac{1}{2})\lambda$ - Destructive interference, $2nt = m\lambda$, where $m = 0, \pm 1, \pm 2, \pm 3, \dots$ e) Explain the application of thin films. <i>*e.g: solar panel, glass tint</i> 7.8 Diffraction by a single slit a) Define diffraction. b) Explain the diffraction of a single slit with the aid of a diagram. c) Use: $y_m = \frac{m\lambda D}{a}$ for dark fringes (minima) $y_m = \frac{(m+\frac{1}{2})\lambda D}{a}$ for bright fringes (maxima), where $m = \pm 1, \pm 2, \pm 3, \dots$ <i>*dark fringes: $m = 1$, 1^{st} dark or 1^{st} order min, $m = 2$, 2^{nd} dark or 2^{nd} order min</i> <i>bright fringes: $m = 1$, 1^{st} bright or 1^{st} order max, $m = 2$, 2^{nd} bright or 2^{nd} order max</i> <i>central bright: Use equation for first dark</i> 7.9 Diffraction grating a) Explain the formation of diffraction with the aid of a diagram. b) Apply $d \sin \theta = m\lambda$ where $d = \frac{1}{N}$ <i>*e.g.: transparent compact disc, muslin cloth, etc.).</i> <i>**bright fringes: $m = 0$, central or 0^{th} order max, $m = 1$, first bright or 1^{st} order max</i> <i>N: number of slits per unit length</i> c) Determine the wavelength of laser beam using a diffraction grating. (Experiment 6: Diffraction Grating) d) Determine the number of diffraction grating lines per unit length. (Experiment 6: Diffraction Grating)
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	Topic 8: Wave Properties of Particle 8.1 de Broglie wavelength - State wave-particle duality. - Use de Broglie wavelength, $\lambda = \frac{h}{p}$ 8.2 Electron diffraction - Describe the observations of electron diffraction in Davisson-Germer experiment. - Explain the wave behaviour of electron in an electron microscope. <i>*relate de Broglie wavelength of electron with the resolving power of the microscope</i> - State the advantages of electron microscope compared to optical microscope.
	Topic 9: Nuclear and Particle Physics 9.1 Binding energy and mass defect - Define and use mass defect, $\Delta m = (Zm_p + Nm_n) - m_{\text{nucleus}}$ - Define and use binding energy, $E_B = \Delta mc^2$ - Determine binding energy per nucleon, $\frac{E_B}{A}$ - Sketch and describe graph of binding energy per nucleon against nucleon number. 9.2 Radioactivity - Explain α, β^+, β^- and γ decays. <i>*radioactive decay as spontaneous and random process</i> <i>**introduce neutrino and antineutrino</i> - State and use decay law $\frac{dN}{dt} = -\lambda N$ - Define and determine activity, A and decay constant, λ. <i>*consider decay curve</i> - Use $N = N_0 e^{-\lambda t}$ or $A = A_0 e^{-\lambda t}$ - Define and use half-life, $T_{\frac{1}{2}} = \frac{\ln 2}{\lambda}$ 9.3 Particle Accelerator - State the thermionic emission. - Explain the acceleration of particle by electric and magnetic field. - State the role of electric and magnetic field in particle accelerators (linac and cyclotron) and detectors (general principles of ionisation and deflection only). - State the need of high energies required to investigate the structure of nucleon. 9.4 Fundamental Particle - Explain the standard quark-lepton model particles (baryons, meson, leptons and hadrons). - Explain the corresponding antiparticle for every particle.

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Panels:

No.	Panel	Institution	Academic Qualification
1.	Assoc. Professor Dr. Rosnah binti Zakaria	Universiti Teknologi MARA	PhD (Material Science), UiTM MScEd (Physics), UKM B. Sc. Ed (Hons) – Physics UKM
2.	Assoc. Professor Ir. Dr Sohiful Anuar bin Zainol Murad	Universiti Malaysia Perlis	PhD (Electronics), Kyushu Univ, Japan MSc (Electronic System Design), USM B.Eng (Electronic), Saga Univ, Japan
3.	En Mohamed Hazri bin Ariffin	Kolej Matrikulasi Kelantan	BScEd (Hons) (Physics), UTM
4.	Pn. Hayati binti Ibrahim	Kolej Matrikulasi Pahang	MEd (Physics), UPSI BSc Edu (Hons) – Physics, UM
5.	En. Nazaruddin bin Zakaria	Kolej Matrikulasi Labuan	MEd (Education Management), UMS BSc with Edu (Physics), UPM
6.	En. Che Mohamad Zamri bin Bakar	Kolej Matrikulasi Perak	BSc (Hons) (Physics), UKM DipEd (Physics), UTM
7.	Pn. Salasiah Binti Hamzah	Kolej Matrikulasi Kejuruteraan Johor	BSc (Physics), UTM DipEd (Physics), MPRM
8.	Pn. Nor Asyiqin Binti Mohd Baharuddin	Kolej Matrikulasi Perlis	BSc (Physics), USM DipEd, MPPPP
9.	En. Elmy bin Abdul Malek	Kolej Matrikulasi Kejuruteraan Pahang	BSc (Hons) (Physics), USM DipEd (Physics), IPGKDRI
10.	Pn. Hasimah binti Husin	Kolej Matrikulasi Kedah	BEng (Kejuruteraan Elektrik & Elektronik), USM DipEd (Physics), MPRM
11.	Pn. Aifaa binti Awang Kechik	Bahagian Matrikulasi KPM	BSc with Edu (Hons) – Physics, UPM
12.	Pn. Jamilah binti Abdul Rashid	Bahagian Matrikulasi KPM	BEng (Hons) Electrical, Electronis & System, UKM
13.	Pn. Siti Aisyah binti Sahdan	Bahagian Pembangunan Kurikulum KPM	BSc. Physics (Hons), University of Nottingham, UK